When Does LVR Outrun Fees? Adverse Selection and Regime Dependence in AMM Liquidity Provision

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Abstract

This note studies whether adverse-selection losses in AMMs scale faster than fee income as market toxicity rises. Using a minute-level panel for a Uniswap v3 WETH/USDC pool on Base matched to an external spot reference price, I construct two LVR-style loss proxies and a regime classification based on rolling measures of volatility, dislocation, and toxicity. The main result is that fee income rises in more toxic regimes, but LVR rises substantially faster, causing net LP economics to deteriorate sharply. Under a harsher swap-markout-based proxy, both non-toxic and toxic regimes are net negative, though toxic regimes are far worse. Under a weaker dislocation-based proxy, non-toxic regimes are modestly positive while toxic regimes remain negative. Across specifications, the directional result is robust: as toxicity increases, LVR scales faster than fees, making adverse-selection-heavy periods materially less attractive for passive liquidity provision.

1. Motivation

The key question in AMM liquidity provision is not whether active markets generate more fees. They usually do. The more important question is whether those fees rise fast enough to compensate LPs for adverse-selection losses. In AMMs, these losses are often summarized as loss-versus-rebalancing (LVR): the cost passive liquidity providers bear when arbitrageurs rebalance stale pool prices against external information. The relevant issue, then, is not simply whether toxic periods are worse, but whether LVR begins to outpace fee growth as toxicity rises.

That scaling question matters because it changes how LP economics should be understood. If fees and LVR grow roughly proportionally, then high-activity environments may still be attractive. But if LVR grows faster than fees, then additional activity can make passive liquidity provision less attractive rather than more. This note asks whether that is what happens in a Base WETH/USDC pool.

2. Data and Construction

I build a minute-level panel for a Uniswap v3 WETH/USDC pool on Base between January 1 and January 7, 2025. For each minute, I record AMM price, swap volume, fee revenue, and swap count, then match the pool to an external spot reference price. From this merged panel I construct dislocation between the AMM and external price, realized volatility, minute-level toxicity measures, and two LVR-style loss proxies.

The first proxy is a harsher swap-markout-based measure that uses the relationship between swap flow and subsequent price movement. The second is a weaker dislocation-based proxy that scales losses using contemporaneous AMM–CEX price differences. I then classify each minute into toxic and non-toxic regimes using rolling threshold rules based on volatility, dislocation, and toxicity. Specifically, the regime indicator is based on one-day rolling threshold rules using the upper tail of these variables. In the resulting sample of 8,640 minutes, approximately 55.4% of minutes are classified as non-toxic and 44.6% as toxic.

Two caveats are important to note. First, these are proxy-based measures of adverse selection rather than exact realized LP PnL. Second, the analysis is conducted at the pool-environment level rather than for a fully simulated LP position with explicit range choices, gas costs, and hedge logic. The goal here is to study whether the environment becomes materially less favorable as toxicity rises, not to claim exact realized returns for a live strategy.

3. Results

The first result is that toxic regimes are more active. Relative to non-toxic periods, toxic periods show higher average volume, higher swap count, higher volatility, and larger AMM–CEX dislocations. This baseline makes intuitive sense, since AMMs are fundamentally short gamma. Average fees rise from about 0.209 USD per minute in non-toxic periods to about 0.513 USD per minute in toxic periods.

However, LVR rises much faster than fees. Under the swap-markout-based proxy, average LVR increases from about 0.249 USD per minute in non-toxic periods to about 1.558 USD per minute in toxic periods. As a result, average net LP economics deteriorate from about -0.040 USD per minute in non-toxic periods to about -1.045 USD per minute in toxic periods. The difference in means is about 1.01 USD per minute, with a t-statistic of about 16.68 and an extremely small p-value. Economically, the result is clear: toxic periods are dramatically worse.

A particularly useful summary statistic is the fee recovery ratio, defined as fees divided by the LVR proxy. In non-toxic periods this ratio is about 0.84, meaning that fees recover roughly 84% of estimated adverse-selection losses. In toxic periods the ratio falls to about 0.33, implying that only about one-third of those losses are recovered through fees. This sharp drop suggests that the regime effect is not just about more trading activity, but about a breakdown in the relationship between activity and compensating fee income. As shown in fig. 1, fee recovery is much weaker in toxic periods.

The toxicity-bucket results strengthen this interpretation. Restricting attention to active minutes with at least one swap, fee income stays relatively stable across quintiles of toxicity while the LVR proxy rises sharply. Net LP economics worsen as toxicity increases, with the most toxic bucket showing by far the most negative outcome. This pattern indicates that the central mechanism is

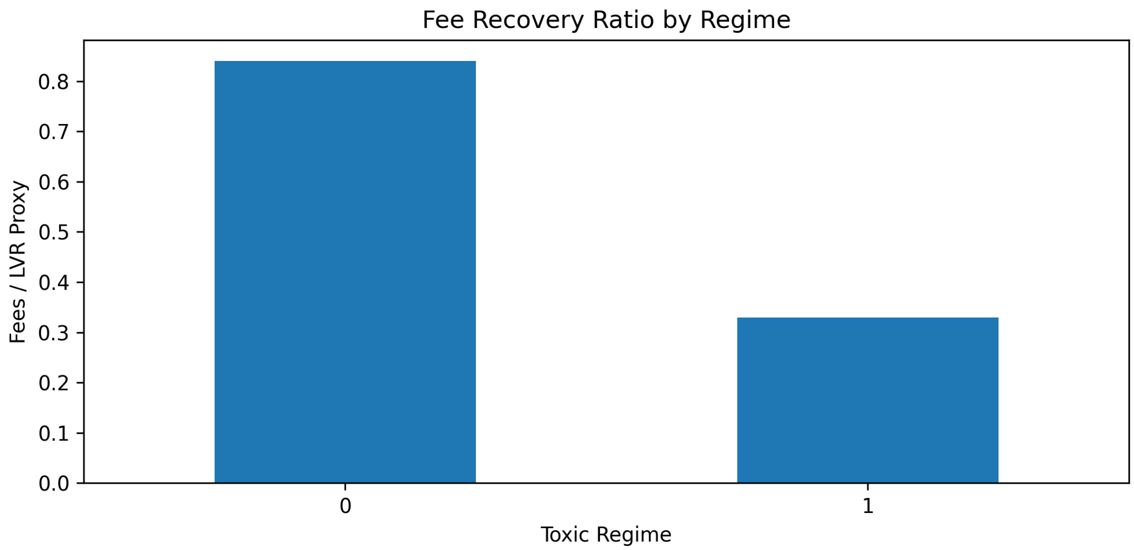

Figure 1: Fee recovery ratio by regime, defined as fees divided by the LVR proxy. Non-toxic periods recover a much larger share of adverse-selection losses through fees than toxic periods.

not a collapse in fee generation, but a much faster increase in adverse-selection cost. As shown in fig. 2, fee income remains relatively stable across toxicity buckets while the LVR proxy rises sharply.

4. Proxy Sensitivity

A useful robustness check is to compare the harsher swap-markout-based proxy with the weaker dislocation-based proxy. Under the weaker proxy, non-toxic periods become modestly positive, with average net economics around 0.081 USD per minute, while toxic periods remain negative at about -0.182 USD per minute. This is important because it shows that the precise profitability conclusion depends on proxy construction, but the relative ranking does not. As shown in fig. 3, the directional ranking is robust across the two loss specifications.

That is the core robust result of the note. Across specifications, non-toxic periods are consistently better than toxic periods, and toxic periods remain economically unattractive. The harsher proxy suggests that even benign environments may fail to fully compensate LPs, while the weaker proxy suggests that benign environments may remain viable. In both cases, the deterioration in toxic periods is large and systematic.

5. Conclusion

The main result of this note is that LP economics are governed by a scaling problem. As market toxicity rises, fee income does increase, but LVR rises substantially faster. Once LVR begins to outpace fee growth, additional activity no longer improves passive liquidity provision and can instead make it materially less attractive.

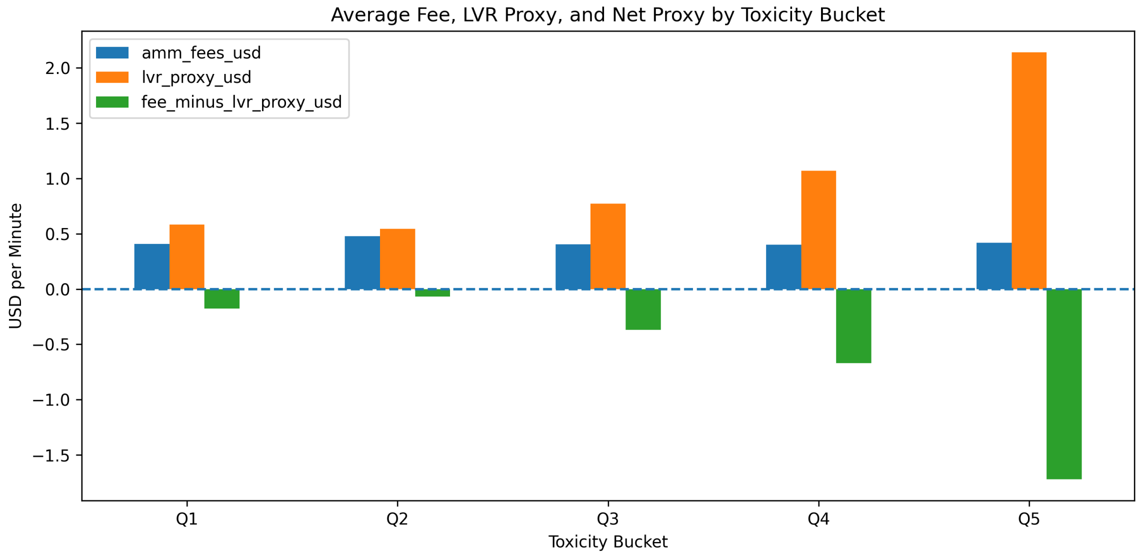

Figure 2: Average fee income, LVR proxy, and net LP proxy across toxicity buckets. Fees remain relatively stable across buckets, while the LVR proxy rises sharply as toxicity increases, causing net LP economics to deteriorate.

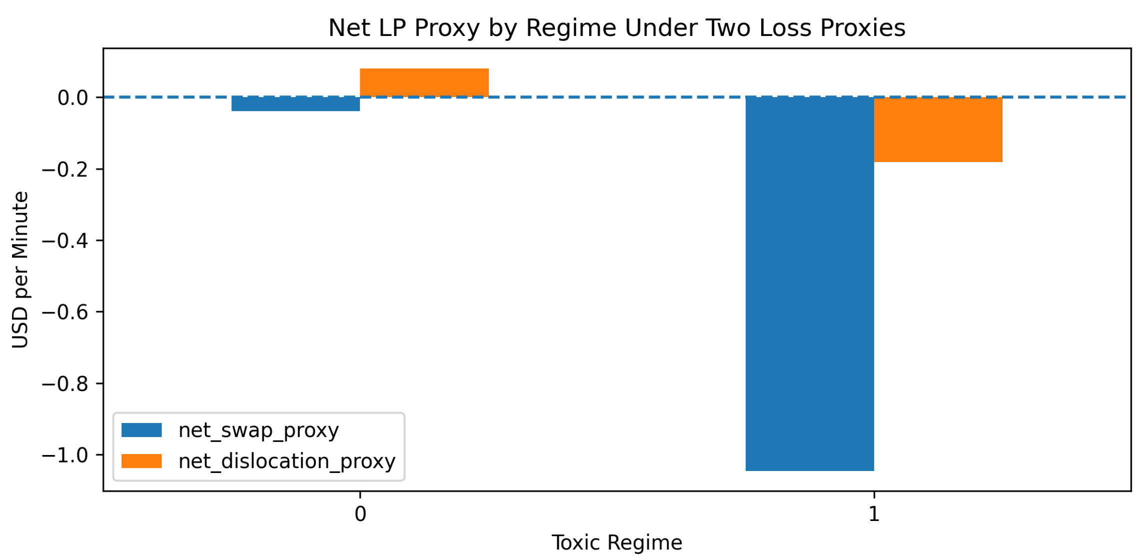

Figure 3: Net LP economics by regime under two loss proxies. The swap-markout-based proxy is harsher, while the dislocation-based proxy is weaker, but both show that toxic regimes are economically worse for LPs.

This has a direct implication for strategy design. A naive always-on LP policy is unlikely to be optimal if the environment shifts between benign and adverse-selection-heavy states. A more promising direction is a regime-aware LP strategy that conditions exposure, range width, or hedging behavior on measures of toxicity, dislocation, and volatility. The next step is therefore to move from pool-level environment analysis to a position-level LP simulation with explicit range rules, intervention costs, and, potentially, a hedge overlay.

Appendix

fig. 4 provides a normalized version of the toxicity-bucket result, while fig. 5 shows the cumulative contribution of toxic and non-toxic periods over time.

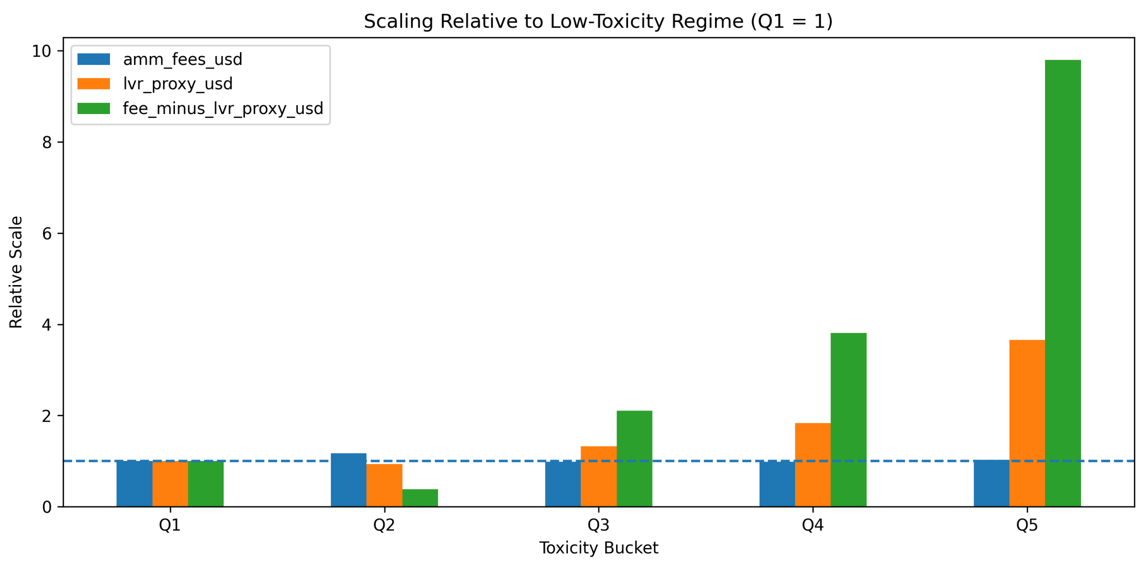

Figure 4: Scaling of fee income, LVR proxy, and net LP proxy relative to the lowest-toxicity bucket (Q1 = 1). The figure shows that LVR grows much faster than fees as toxicity rises.

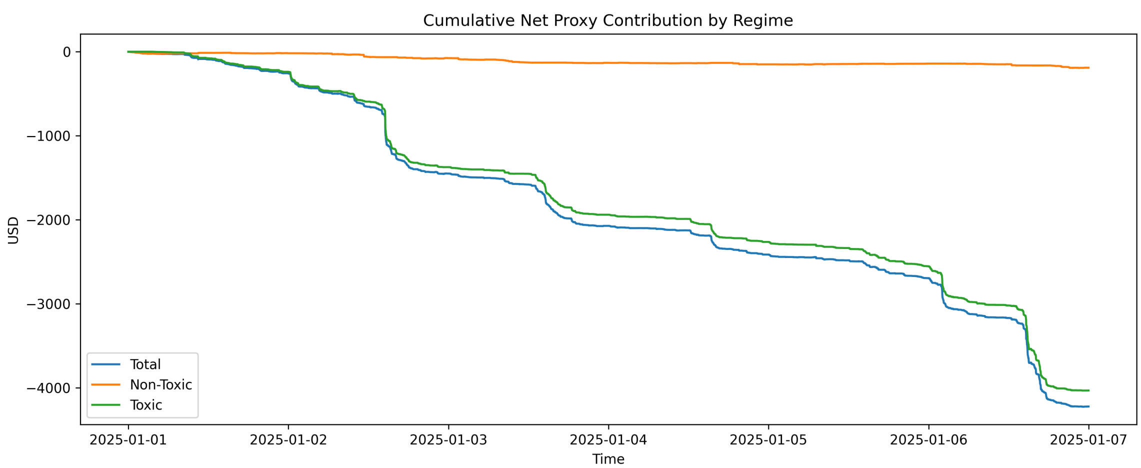

Figure 5: Cumulative net LP proxy contribution by regime over time. Toxic periods contribute disproportionately to the deterioration in cumulative LP economics.